

VISHNU BHUVANACHANDRAN

Mechanical Project Engineer | Environmental Safety & Occupational Health | Project Delivery | Site H&S
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PROFESSIONAL SUMMARY

Mechanical Engineer with a PhD specialising in Environmental Safety and Occupational Health, supported by extensive experience in mechanical engineering project coordination, subcontractor management, technical documentation, and Health & Safety compliance. Proven track record of managing funded engineering projects from initiation through to completion, coordinating multidisciplinary teams, and ensuring all works are carried out in line with Health & Safety requirements. Experienced in quality management systems, technical submittals, progress reporting, and procurement coordination. Based in Dundalk, Co. Louth immediately available and ideally located for County Louth site-based projects. Holds a full driving licence.

KEY SKILLS

Mechanical Engineering: Thermal systems, fluid mechanics, process engineering, mechanical installations, pipework principles

Health & Safety: Environmental safety, occupational health (PhD specialism), safe systems of work, NIOSH & AWS compliance, hazard identification and control

Project Coordination & Delivery: Multi-site project delivery, programme milestone monitoring, subcontractor coordination, site activity management

Technical Documentation: Drawing review, specifications, technical submittals, quality inspection records, site records, progress reports

Procurement & Materials: Budget tracking, material management, vendor coordination, actual vs. budget variance reporting

Quality Management: Quality inspections, testing support, commissioning and handover procedures, governance compliance

Stakeholder Liaison: Project managers, consultants, client representatives, multidisciplinary engineering teams

PROFESSIONAL EXPERIENCE

Marian Engineering College (APJ Abdul Kalam Technological University)

Kerala, India

Research Lead, Associate Professor & Innovation Coordinator — Mechanical Engineering

Sept 2019 – Aug 2024

- Managed full project lifecycle delivery for a funded EV Battery Cooling System project (INR 12 Lakhs / ~EUR 13,000): coordinated subcontractor-equivalent specialist teams, managed procurement and material tracking, produced technical documentation, and monitored programme milestones from design through to testing and commissioning.
- Coordinated a 30+ member interdisciplinary engineering team on the Electric Go-Kart infrastructure project, managing daily site activities, scheduling, quality inspections, and delivery against programme milestones — achieving national-level recognition.
- Served as Assistant Nodal Officer of the Innovation and Entrepreneurship Development Centre (IEDC) and Convener of the Institute Innovation Council (IIC), overseeing institutional compliance, governance frameworks, and cross-functional engineering initiatives.
- Led an INR 28 Lakhs (~EUR 30,000) government-funded Cluster Mapping project: managed stakeholder liaison with client representatives and consultants, maintained full quality management documentation, and produced structured progress reports throughout delivery.
- Produced 10+ peer-reviewed publications and secured 5 Indian patents, demonstrating rigorous technical documentation, engineering problem-solving, and quality management discipline across complex, multi-phase engineering projects.

Indira Gandhi Centre for Atomic Research (IGCAR) — Dept. of Atomic Energy, Govt. of India

India

Senior Research Fellow — Environmental Safety & Occupational Health

May 2014 – Jan 2019

- Designed and executed experimental protocols for industrial emission characterisation and process control within one of India's most strictly regulated engineering environments, maintaining safe systems of work in full compliance with NIOSH and AWS international Health & Safety standards.
- Achieved a 70% reduction in hazardous Cr(VI) emissions through systematic data-driven process improvements — a direct, quantified Health & Safety and environmental compliance outcome across live industrial operations.
- Produced structured compliance and performance reports for health and safety oversight panels; ensured all technical documentation met regulatory and quality management system requirements.
- Coordinated with cross-functional engineering, chemistry, and safety teams to identify and resolve technical issues during live process operations, achieving a 90% improvement in process efficiency and stakeholder confidence.
- Maintained strict data accuracy and governance records across all experimental phases — equivalent to the site records, quality inspection documentation, and reporting disciplines required in construction project delivery.

EDUCATION

National Institute of Technology Tiruchirappalli (NIT-T), India <i>PhD in Mechanical Engineering — Environmental Safety & Occupational Health DAE-IGCAR Government Fellowship</i>	2014 – 2019
Dundalk Institute of Technology, Ireland <i>MSc in Data Analytics — First Class Honours with Gold Medal</i>	2024 – 2025
University of Kerala, India <i>M.Tech in Thermal Science B.Tech in Mechanical Engineering</i>	2011 – 2013 2005 – 2009

KEY ACHIEVEMENTS

- PhD in Environmental Safety & Occupational Health (NIT-T, India) — specialist academic qualification directly aligned to the site H&S responsibilities of this role.
- International speaker at two IIW (International Institute of Welding) General Assemblies — a globally recognised forum for mechanical and welding engineering professionals spanning 50+ countries.
- 5 Indian Patents and 10+ peer-reviewed publications across mechanical engineering, process design, and environmental safety.
- Gold Medal — MSc Data Analytics, First Class Honours, Dundalk Institute of Technology, Ireland (2025).
- Multiple competitive DAE-IGCAR Government of India research fellowships from the Department of Atomic Energy.
- 70% reduction in Cr(VI) hazardous emissions through systematic process engineering and compliance-led intervention.